VII SEMESTER (OPEN ELECTIVE III)	L	T	P	C
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20ME7O01 - BIO-MECHANICAL ENGINEERING				

COURSE OBJECTIVES:
The main objectives of this course are
1. To make the student familiar with fundamentals of bio mechanics.
2. To gain knowledge about musculoskeletal system.
3. To impart knowledge about linear kinetics and angular kinetics
4. Make the student to illustrate the mathematical models used in the analysis of bio-mechanical systems

COURSE OUTCOMES:
After completion of this course, the students are able to,
CO1: Explain about fundamentals of Bio mechanics. [K2]
CO2: Describe the mechanics of musculoskeletal system. [K2]
CO3: Relate the concept of kinetics with human motion. [K3]
CO4: Explain mechanical analysis of human motion. [K3]
CO5: Analyze human movements. [K4]

UNIT-I:
INTRODUCTION TO BIO MECHANICS
Principles of mechanics in human movement, Qualitative and quantitative Analysis, Key mechanical concepts of mechanics and basic units, Nine fundamentals of biomechanics, Nine principles for application of Biomechanics.

UNIT-II:
MECHANICS OF MUSCULOSKELETAL SYSTEM
Principles of joint motions, Muscle structures, Mechanical method of muscle action analysis, Tissue loads and forces, Biomechanics of bones and ligaments, Three mechanical characters of muscle, stretch-shortening cycle (SSC).

UNIT-III:
LINEAR KINETICS AND ANGULAR KINETICS
Vector analysis of angle of pull and muscle angle pull, Contact forces, Impulse-Momentum Relationship, Force-Time Principle, Work-Energy relationship, Segmental interaction principle, Torque, Equilibrium, Center of gravity and Principle of balance.

UNIT-IV:
MECHANICAL ANALYSIS OF HUMAN MOTION
Linear kinematics - linear kinematic analysis, position and displacement, velocity and speed, acceleration, differentiation and integration, kinematics of running, kinematics of projectiles, equations of constant acceleration, Angular kinematics - angular motion, measurements of angles, types of angles, representation of angular motion vectors, lower extremity joint angles, relationship between angular and linear motion, angular kinematics of running.

UNIT-V:
APPLICATIONS OF MEDICAL REHABITATION
Qualitative analysis of kicking technique, batting, catching, throwing techniques, injury risk assessment, equipment design for strength training, Injury mechanics, injury prevention.

Text Books:

1. Ronald L. Huston, Principles of Biomechanics,1st edition CRC Press, 2019.
2. Joseph E. Muscolino, " Kinesology", 3rd edition, Mosby, 2016.
3. Subrata Pal, "Textbook of Biomechanics", 1st edition, Springer US, 2016.

Reference Books:

1. Duane Knudson, "Fundamentals of Biomechanics", 2nd edition, Springer, 2013.
2. Ajay Bahl, "Basics of Biomechanics", 1st edition, Jaypee Brothers Medical Publishers, 2010.
3. Robert frost, "Applied Kinesiology", 1st edition, North Atlantic Books, 2013.
4. David A. Winter, "Biomechanics and Motor Control of Human Movement", John Wiley & sons, 2009.

Web Reference:

1. https://archive.nptel.ac.in/courses/112/105/112105305/
2. https://archive.nptel.ac.in/courses/112/106/112106248/